



Fundus GLAUCOMA Detection using CNN

Computer Aided Engineering



Outline :

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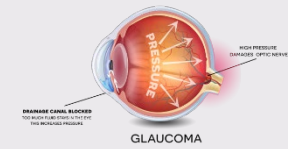


Introduction

Glaucoma Disease



NORMAL EYE



GLAUCOMA



Glaucoma is an **eye disorder** that within **the eyes yields intraocular pressure** (IOP) which damage the **Optic Nerve**.

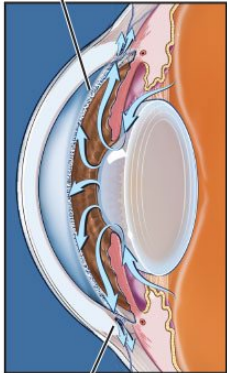
There are two type of Glaucoma : Open-Angle Glaucoma and Closed-Angle Glaucoma



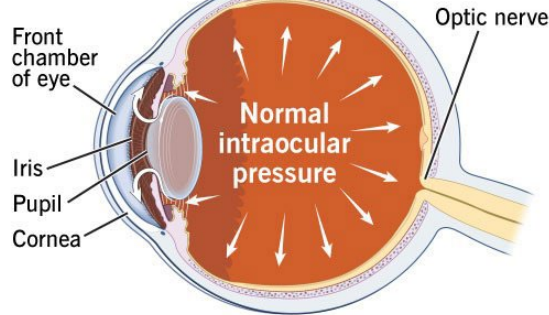
Glaucoma

Healthy Eye

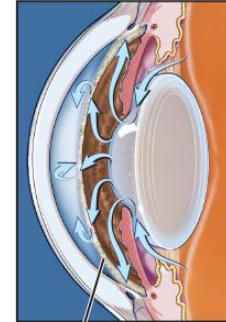
Drainage canal of eye



Aqueous humor leaving through canal

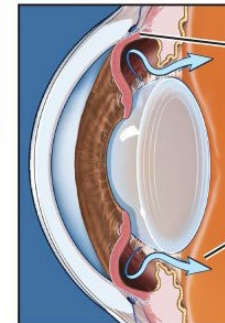


Open-angle glaucoma



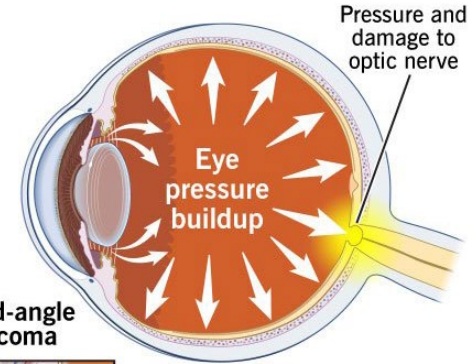
Drainage canals are clogged, fluid pressure builds up

Closed-angle glaucoma



Angle between the iris and cornea is too narrow

Fluid pressure builds up



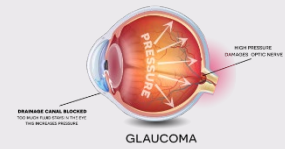


Introduction

Glaucoma Disease



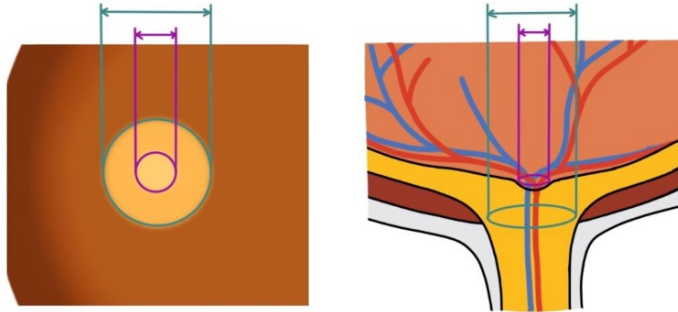
NORMAL EYE



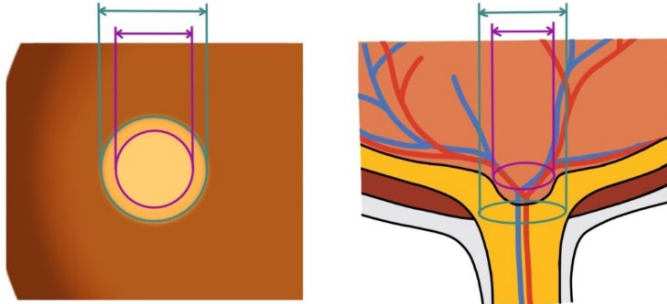
GLAUCOMA

Optic Nerve which damage by Glaucoma

NORMAL OPTIC DISC AND CUP

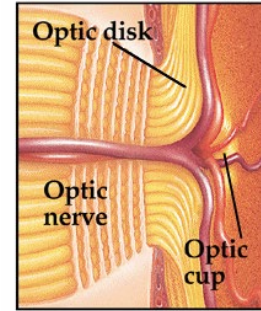


CUPPING IN GLAUCOMA

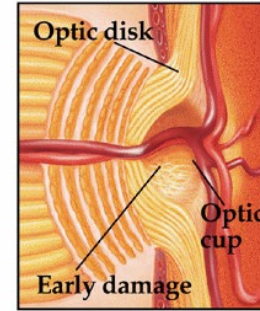


Human Optic Nerve

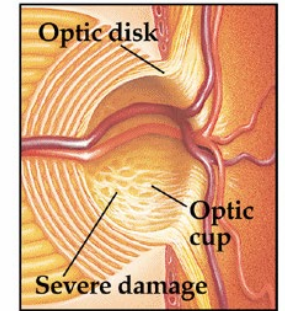
Normal optic nerve



Early glaucoma



Advanced glaucoma



Healthy eyes



Peripheral vision loss due to glaucoma

Human Vision

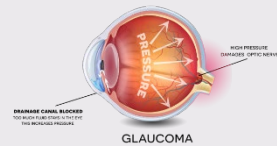


Introduction

Unmeet Need



NORMAL EYE



GLAUCOMA



Currently, the most ways to detect **Glaucoma** is using **Dilated eye exam** with Fundus Camera to get the fundus image.



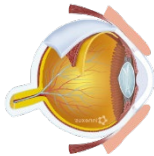
Then the **clinician** needs to identify the boundary contours of the cup and disc and classify into **Positive** or **Negative** Glaucoma.



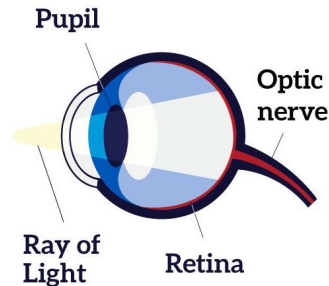
Glaucoma detection is a challenging and lengthy process, relying on multiple examinations and clinical expertise.

Objective :

This study aim to develop **Negative Fundus Glaucoma** vs **Positive Fundus Glaucoma** using **Convolutional Neural Network (CNN)**.



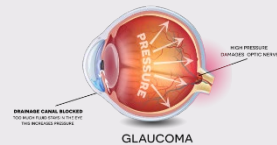
Dilated pupil



Portion of the retina that can be seen through a dilated pupil.

Datasets & Method

Datasets of Fundus Glaucoma Images



The dataset is from :

<https://www.kaggle.com/datasets/sabari50312/fundus-pytorch>

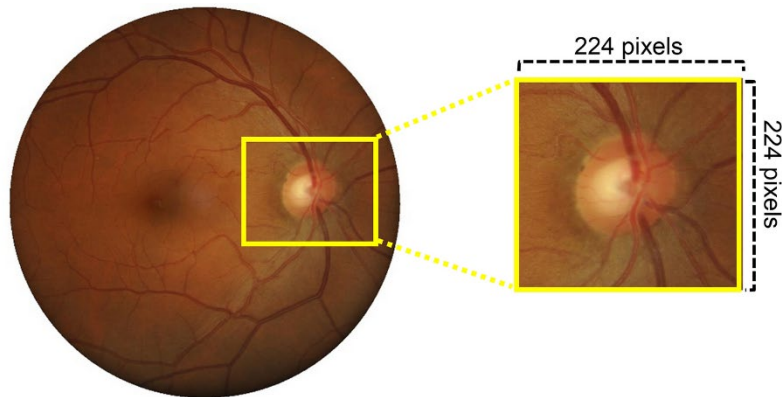
The number of Dataset

7170 Images

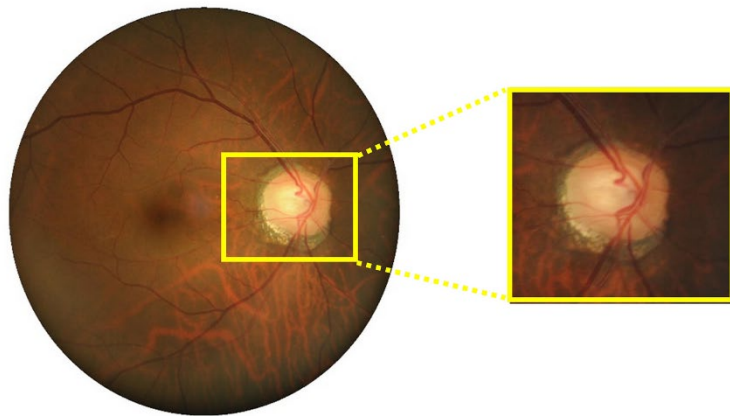
Training and Validation Dataset
(**3440 Negative** vs **3530 Positive**)

Testing Dataset
(**100 Negative** vs **100 Positive**)

Normal

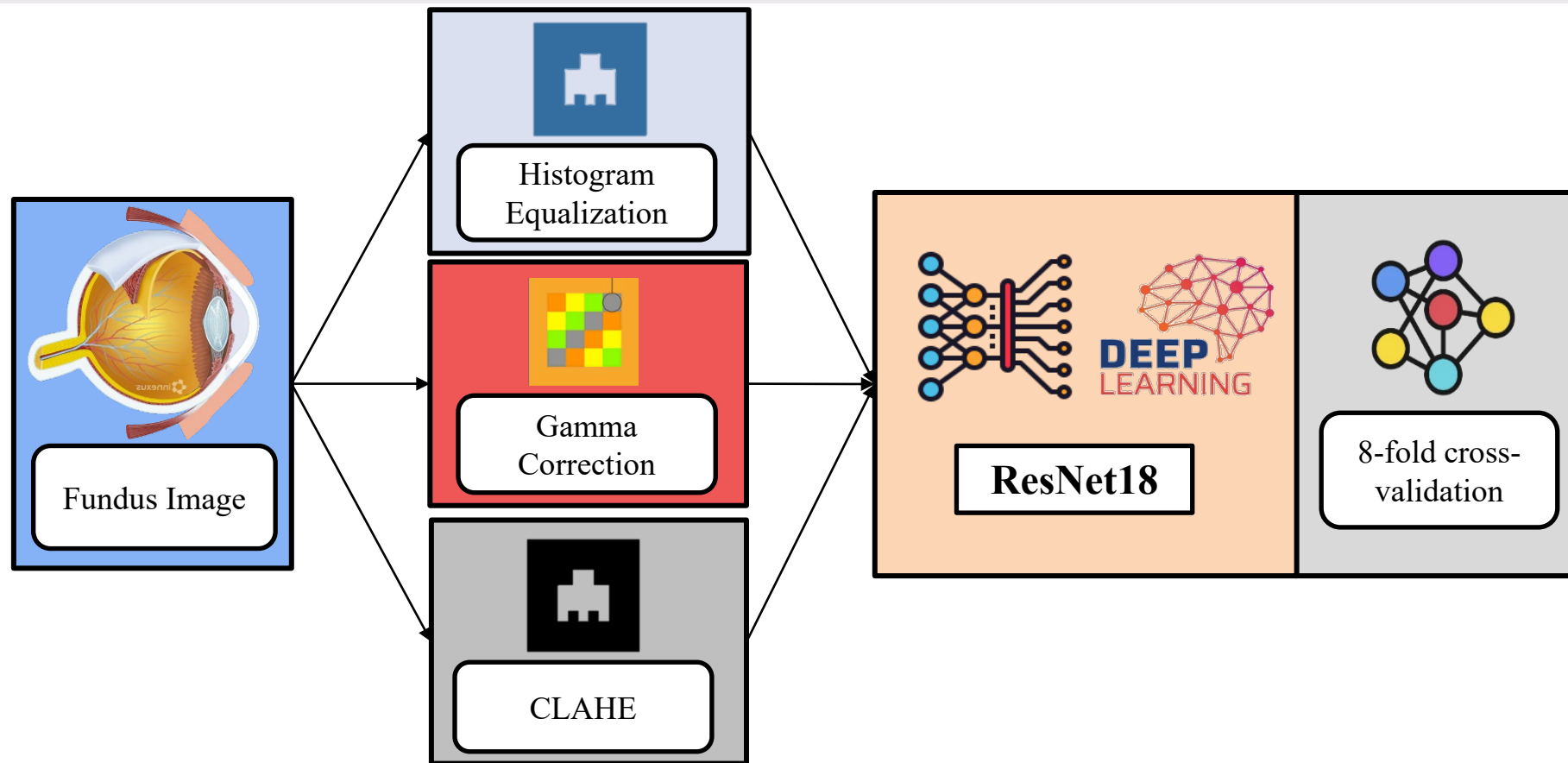
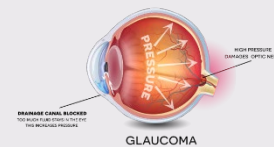


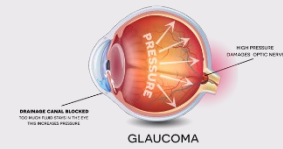
Glaucoma



Datasets & Method

Flow Chart

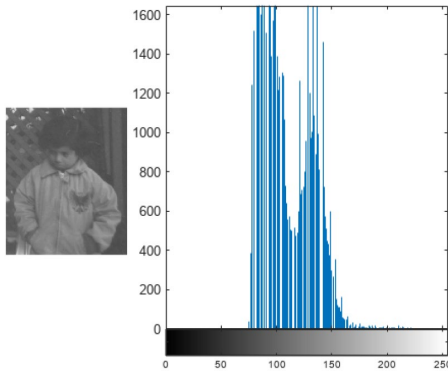




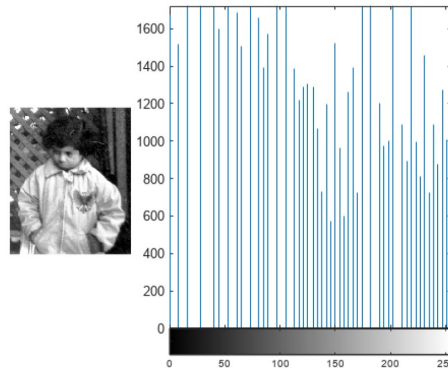
Histogram Equalization

Histogram equalization accomplishes this by **effectively spreading out the most frequent intensity values in a Histogram**. This allows for areas of lower **local contrast to gain a higher contrast**.

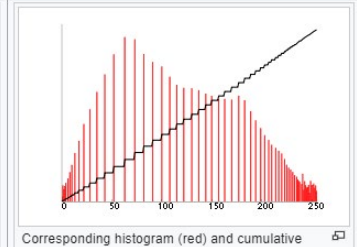
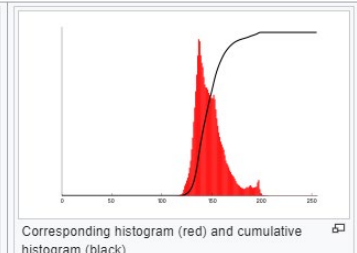
In Matlab to gain this IM method, with code `"Img = histeq(your_image)"`.

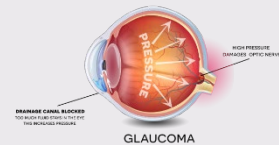


Original Image



IM Image





Gamma Correction

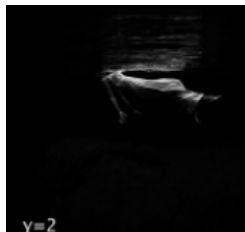
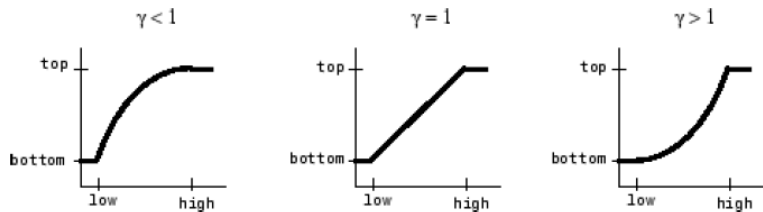
Gamma Correction, gamma can be any value between 0 and infinity. It is used to adjust the luminance of an image.

The goal of this technique is to **alternates the pixel value to improve the image**.

Gamma is 1 (default) : the mapping is linear

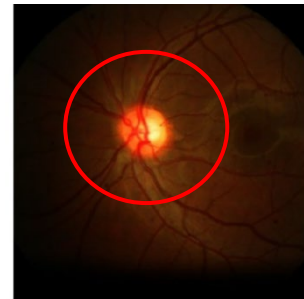
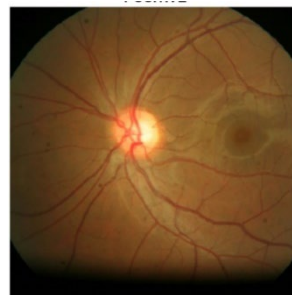
Gamma is less than 1 : the mapping is weighted toward higher (brighter)

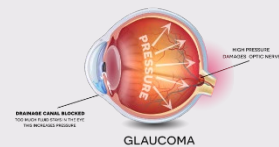
Gamma is greater than 1: the mapping is weighted toward lower (darker)



In Matlab to gain this method, with code:

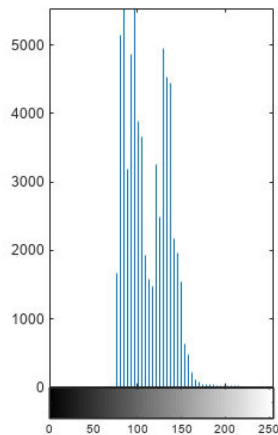
```
Img = lin2rgb(your_image);
```



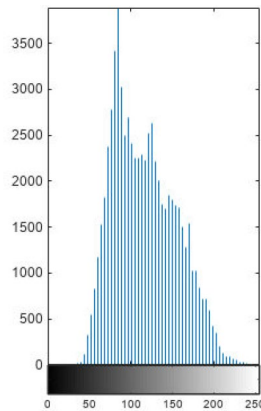


CLAHE

CLAHE or **contrast-limited adaptive histogram equalization**. This is the variant of Adaptive histogram equalization (AHE) that limits contrast amplification to **reduce noise amplification**. While histeq works on the entire image, CLAHE operates on **small regions in the image** called tiles.



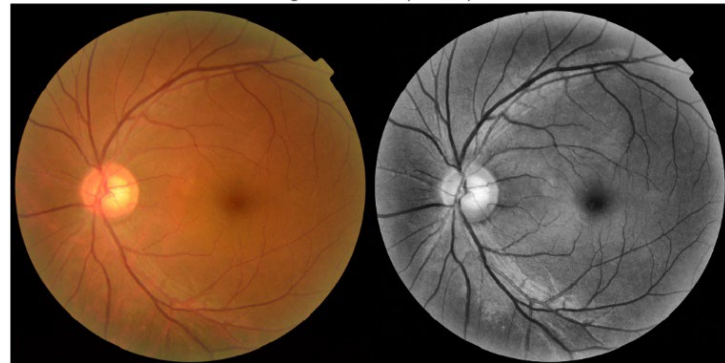
Original Image



IM Image

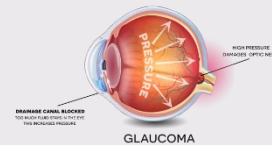
In Matlab to gain this method, with code:
"img = adapthisteq (your_image)".

Original vs CLAHE (Positive)



Datasets & Method

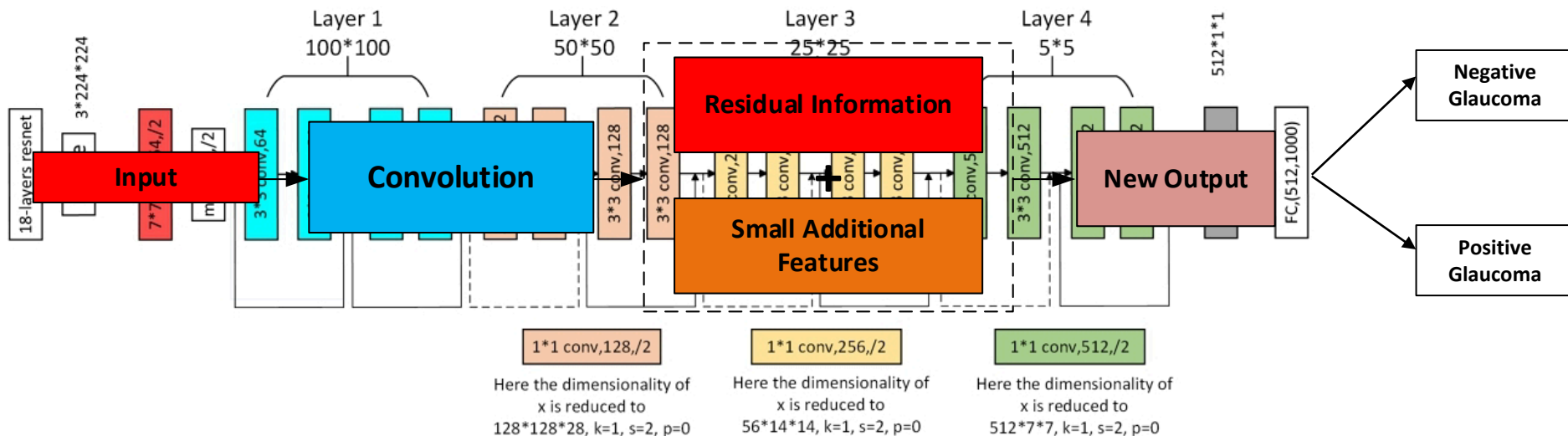
Deep Learning Architecture



A **Convolutional Neural Network (ConvNet/CNN)** is a Deep Learning algorithm which can take in an **input image**, **assign importance** (learnable weights and biases) to various aspects/objects in the image and be able **to differentiate one from the other**. (Saha,2018)

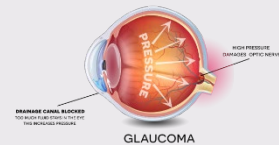
ResNet-18 known as Residual Network. The number 18 denote the number of layers and compute as follows:

1 +(4+4+4+4)+1. It architecture comprise of what we called Residual Blocks. The basic building block for ResNets are the conv and identity blocks.



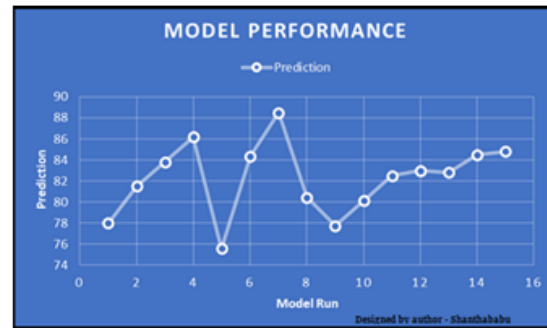
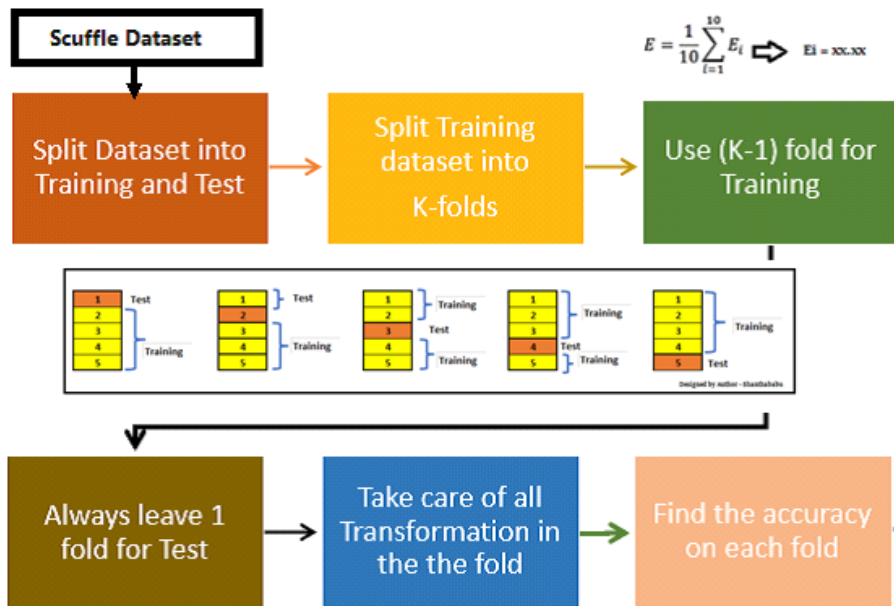
Datasets & Method

Cross Validation (8-fold Cross Validation)



K-fold Cross Validation

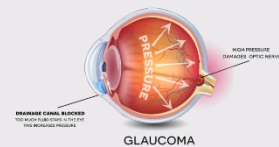
The data set is **split into a K number** of sections/**folds** where **each fold** is used as a **testing set at some point**.



K Fold Cross Validation

Datasets & Method

Evaluation Parameters



Confusion Matrix		Actual Class	
		A	B
Predicted Class	A	TP	FP
	B	FN	TN
Total		P	N

Note:

A = Condition Positive set

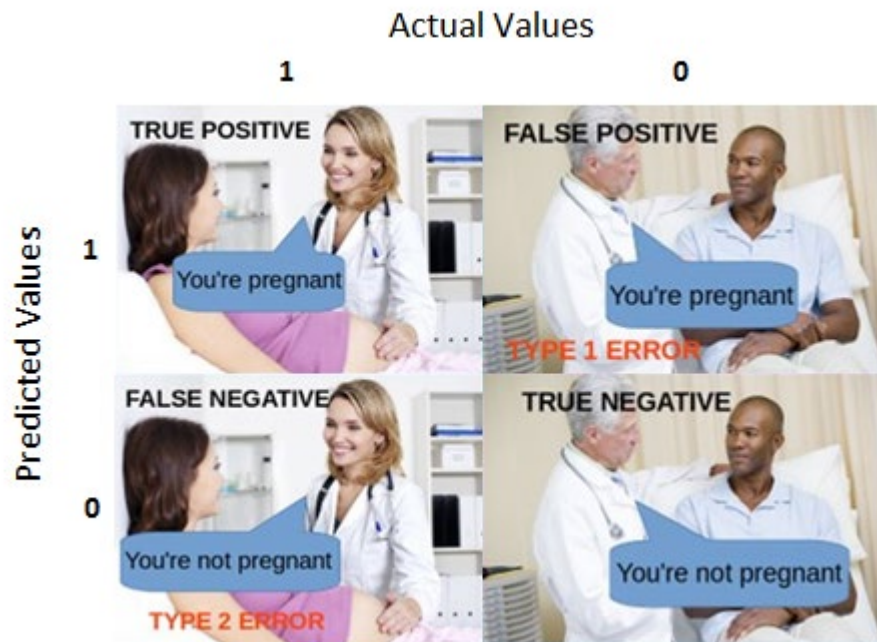
B = Condition Negative set

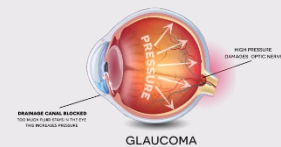
TP = True Positive

FN = False Negative

: FP = False Positive

: TN = True Negative





Confusion Matrix		Actual Class		<i>Sensitivity</i> $= \frac{TP}{TP + FN}$	<i>Specificity</i> $= \frac{TN}{TN + FP}$
		A	B		
Predicted Class	A	TP	FP		
	B	FN	TN		
Total		P	N	<i>Accuracy</i> = $\frac{TP + TN}{P + N}$	

Note:

A = Condition Positive set

B = Condition Negative set

TP = True Positive

FN = False Negative

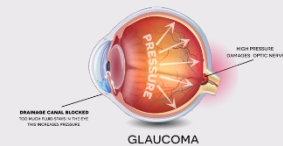
: FP = False Positive

: TN = True Negative

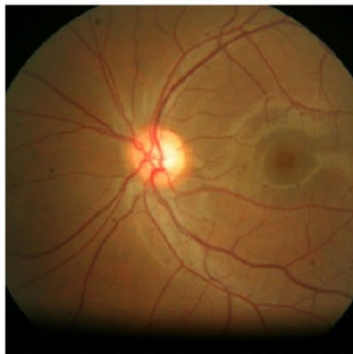


Results

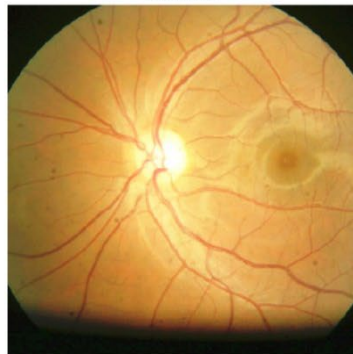
Image Processing Result



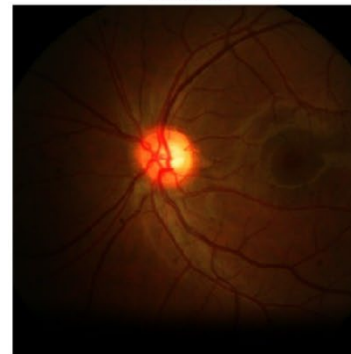
Original class is:
POSITIVE



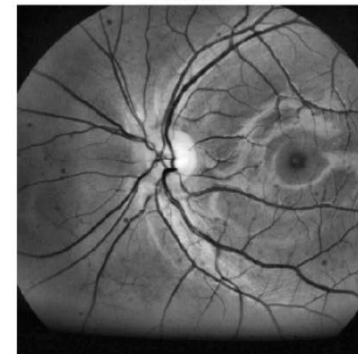
Histogram Equalization class is:
POSITIVE



Gamma Correction class is:
POSITIVE



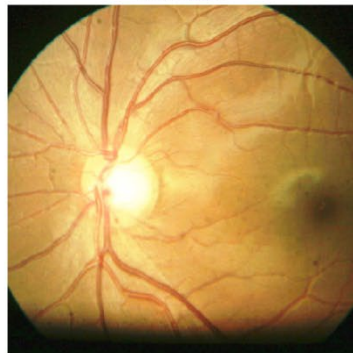
CLAHE Correction class is:
POSITIVE



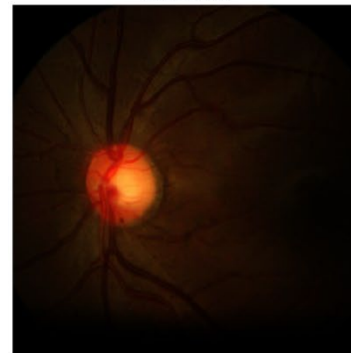
Original class is:
NEGATIVE



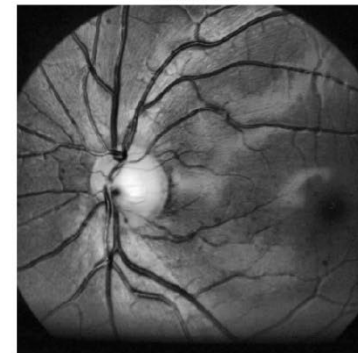
Histogram Equalization class is:
NEGATIVE

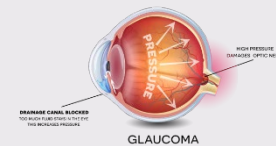


Gamma Correction class is:
NEGATIVE

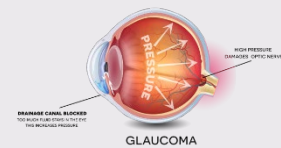


CLAHE Correction class is:
NEGATIVE



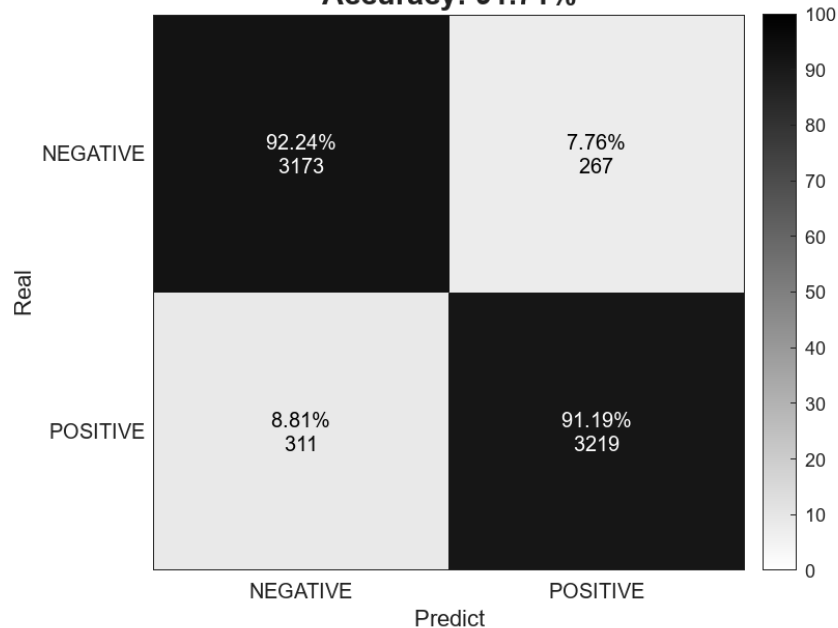


Evaluation Parameter	Image Types			
	Original	Histogram Equalization	Gamma Corrections	CLAHE
Accuracy (%)	91.707	90.645	91.334	90.469
Sensitivity (%)	91.714	90.644	91.344	90.466
Specificity (%)	91.714	90.644	91.344	90.466



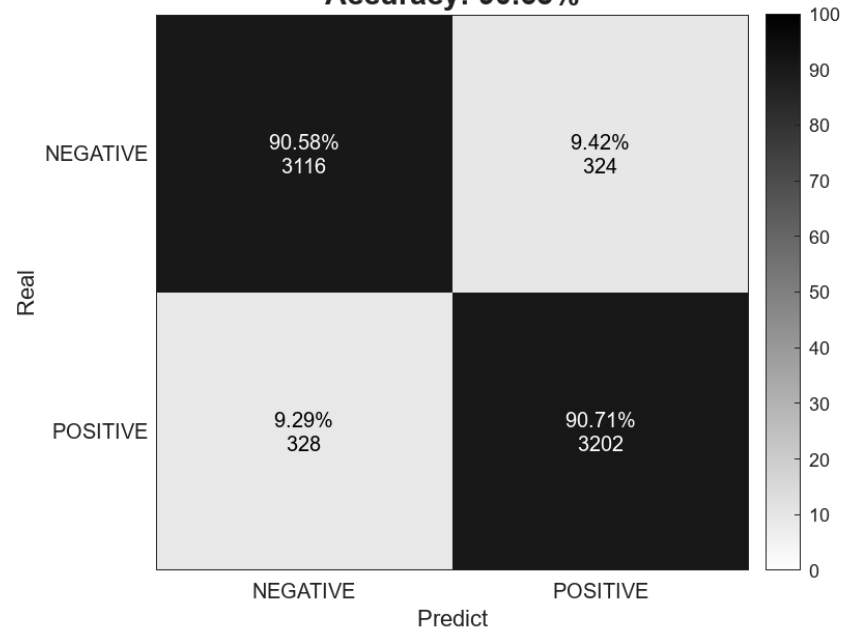
ORIGINAL

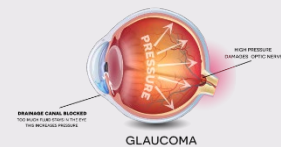
Accuracy: 91.71%



HISTOGRAM

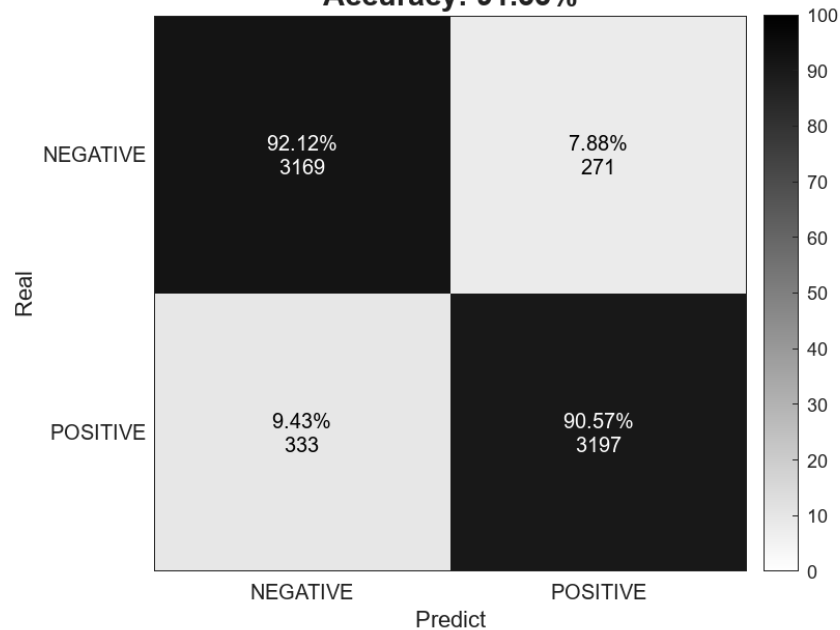
Accuracy: 90.65%





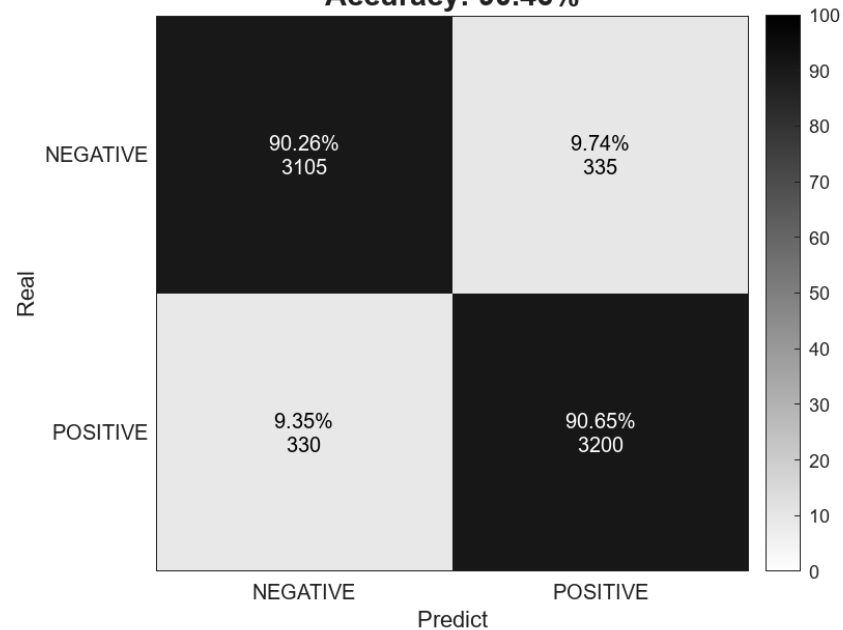
GAMMA

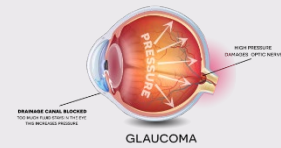
Accuracy: 91.33%



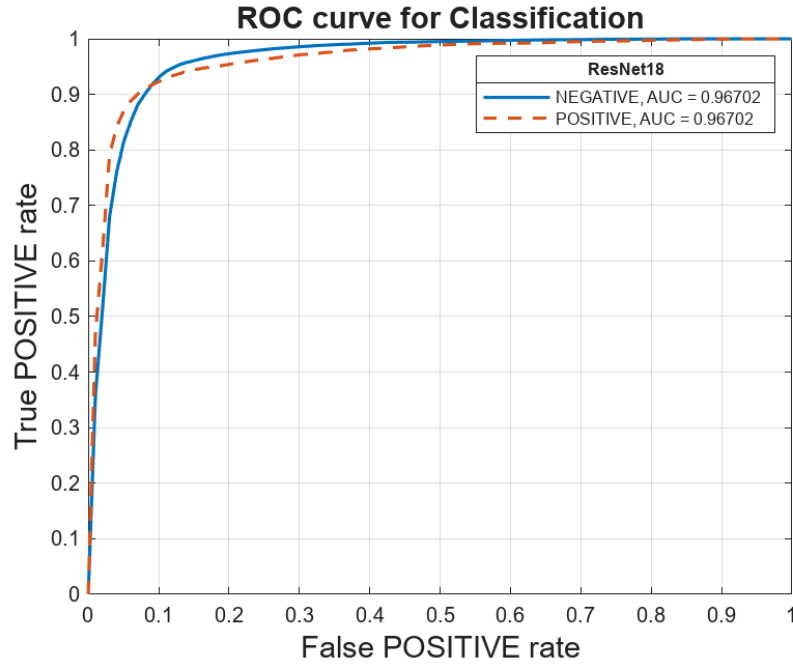
CLAHE

Accuracy: 90.46%



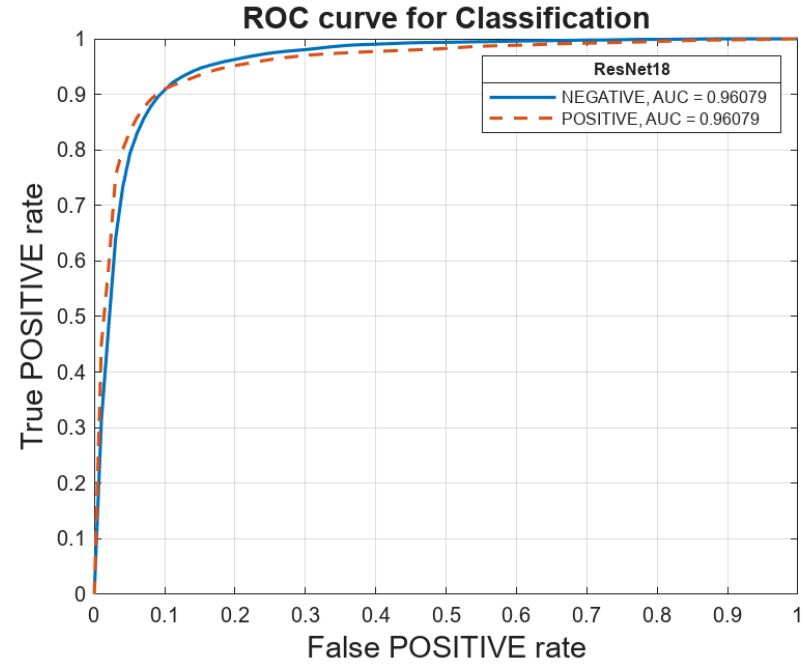


ORIGINAL



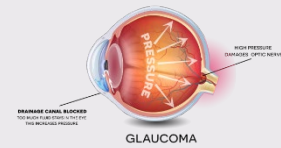
AUC = 0.96702

HISTOGRAM

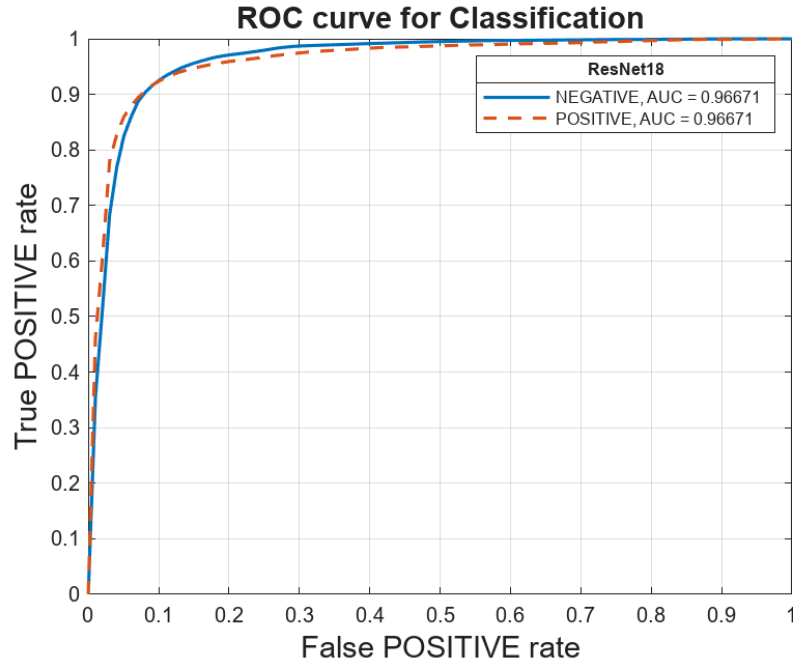


AUC = 0.96079

AUCs (Area Under The ROC Curve) are close to unity. This means that the model is capable of distinguishing between classes

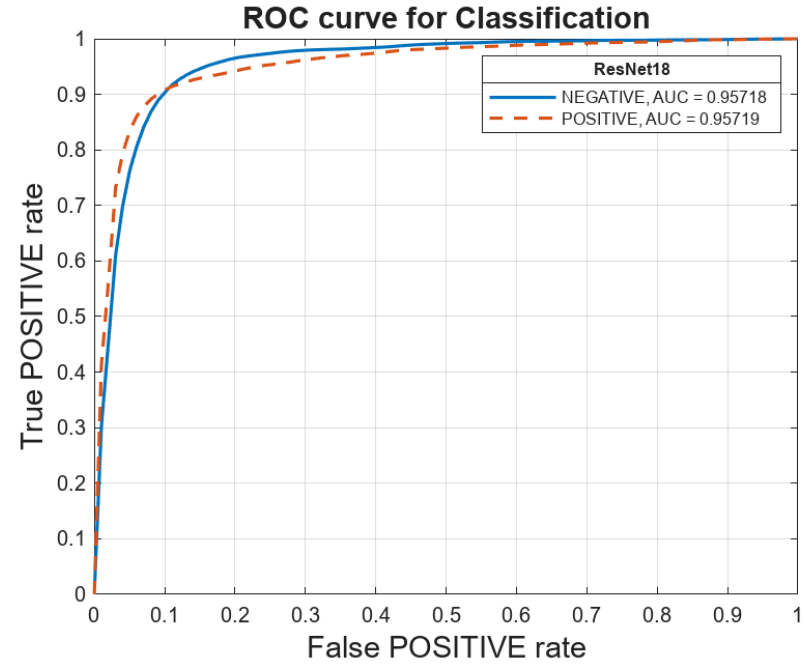


GAMMA



AUC = 0.96671

CLAHE

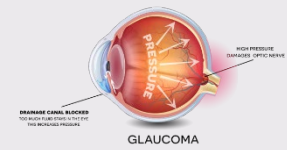


AUC = 0.95719

AUCs (Area Under The ROC Curve) are close to unity. This means that the model is capable of distinguishing between classes

Discussion

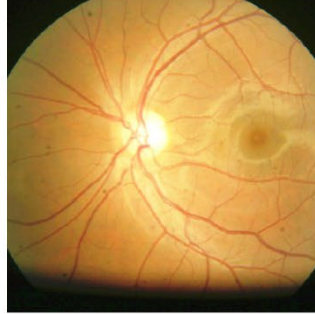
Visualization Classification Image with Score-CAM



Original class is:
POSITIVE



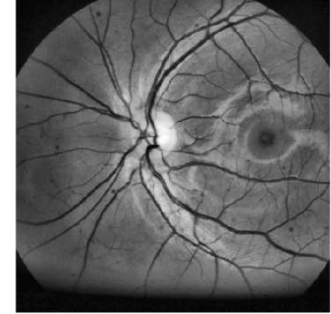
Histogram Equalization class is:
POSITIVE



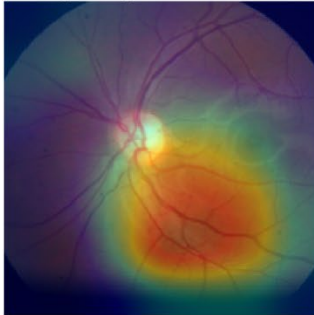
Gamma Correction class is:
POSITIVE



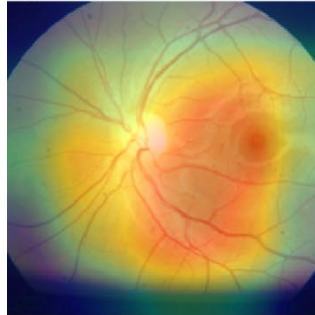
CLAHE Correction class is:
POSITIVE



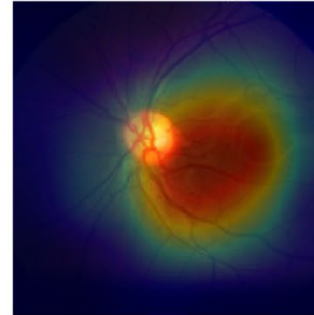
Original Image Fundus Score-CAM



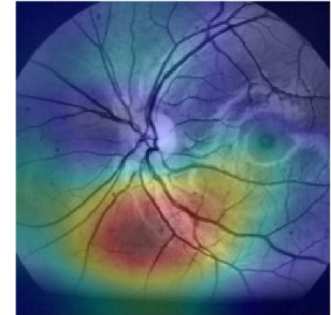
Histogram Equalization Fundus Image Score-CAM



Gamma Correction Fundus Image Score-CAM



CLAHE Correction Fundus Image Score-CAM

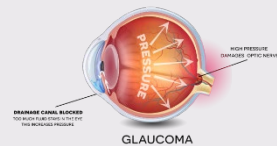


Conclusion

Fundus GLAUCOMA Detection using CNN



NORMAL EYE



GLAUCOMA

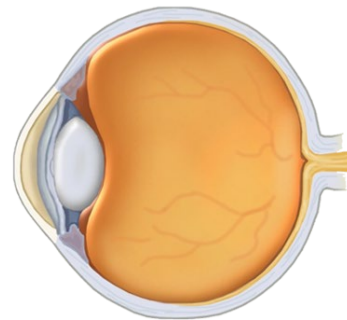


This project proposes the use of **Fundus Glaucoma Images** as an input to a CNN for **classifying Negative and Positive Glaucoma**.

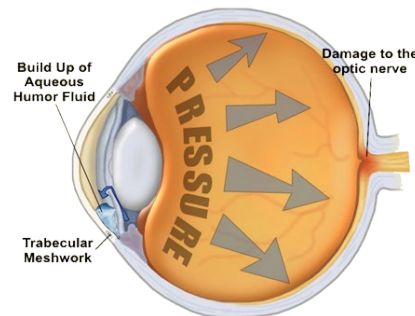


With an **accuracy of 91.71%** with **original image model**, the **ResNet18** model is **achieved the high performance**.

Normal Eye

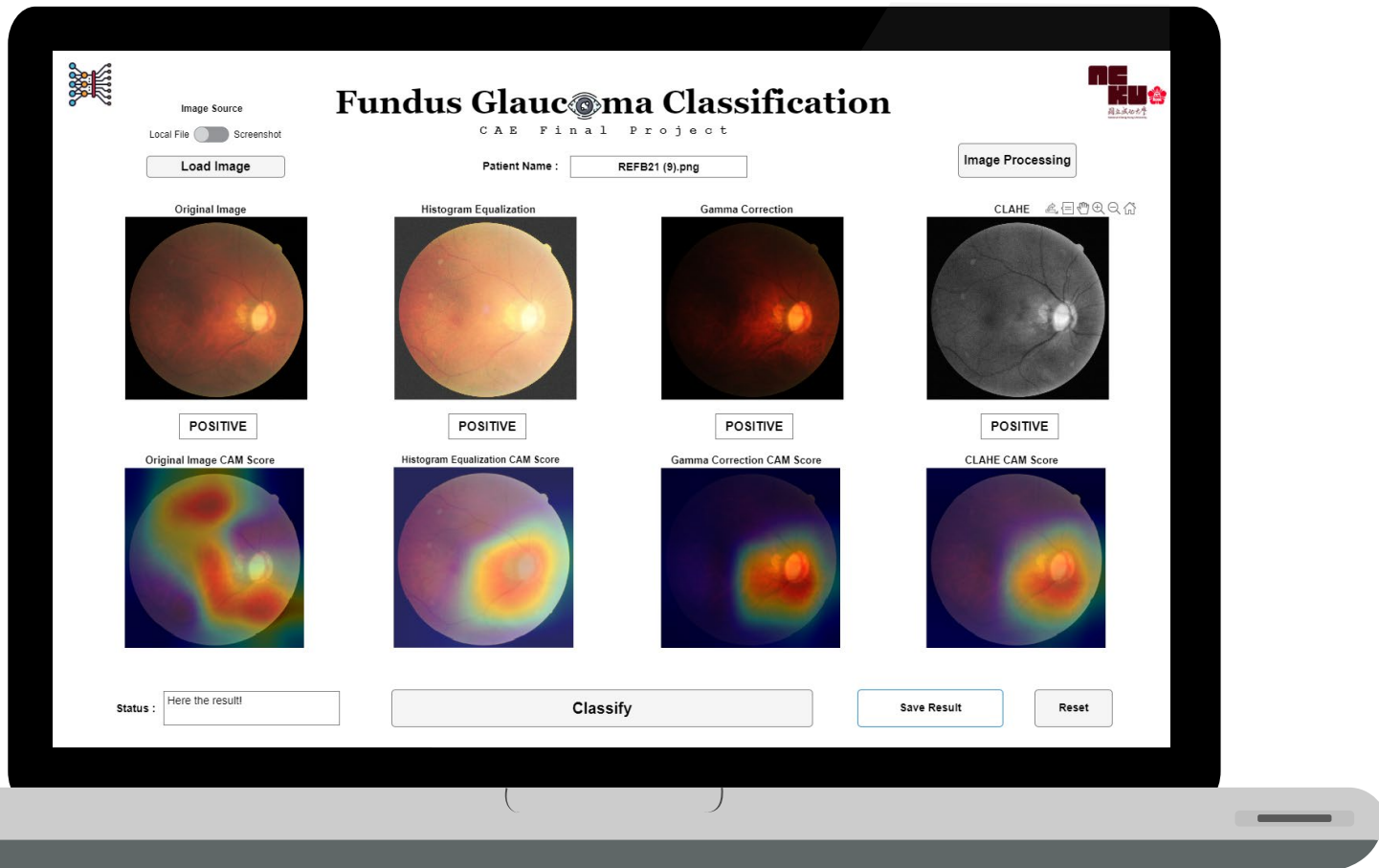


Eye with Glaucoma



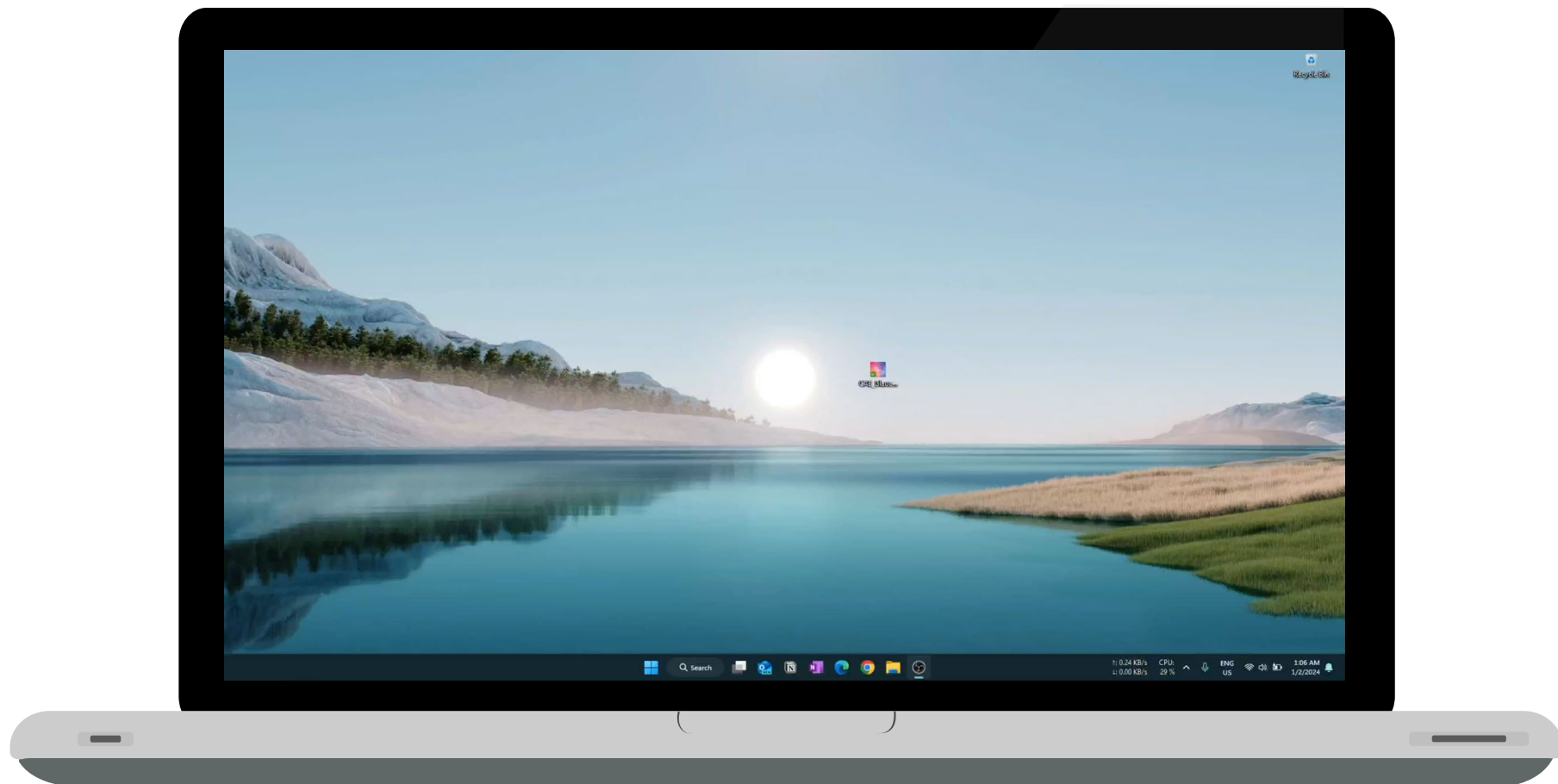


App GUI

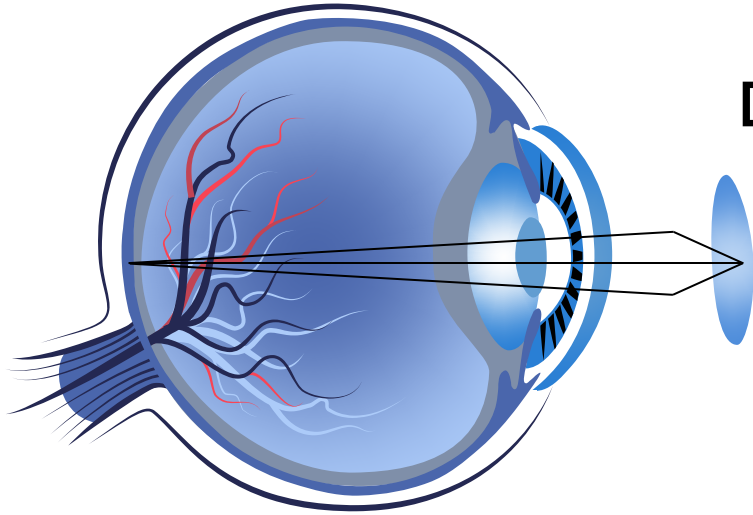




App Demo



THANKS!



DO YOU HAVE ANY QUESTIONS?

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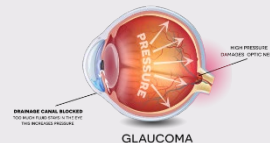


Reference:

All



NORMAL EYE



GLAUCOMA



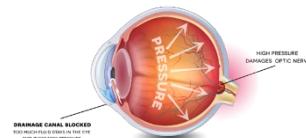
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Conclusion

Convolutional Neural Network
based on **color fundus image** can
classify **Glaucoma** with best
accuracy of 91.71% on Original
Training Images.



NORMAL EYE



GLAUCOMA



App Demo

